CE 400/500

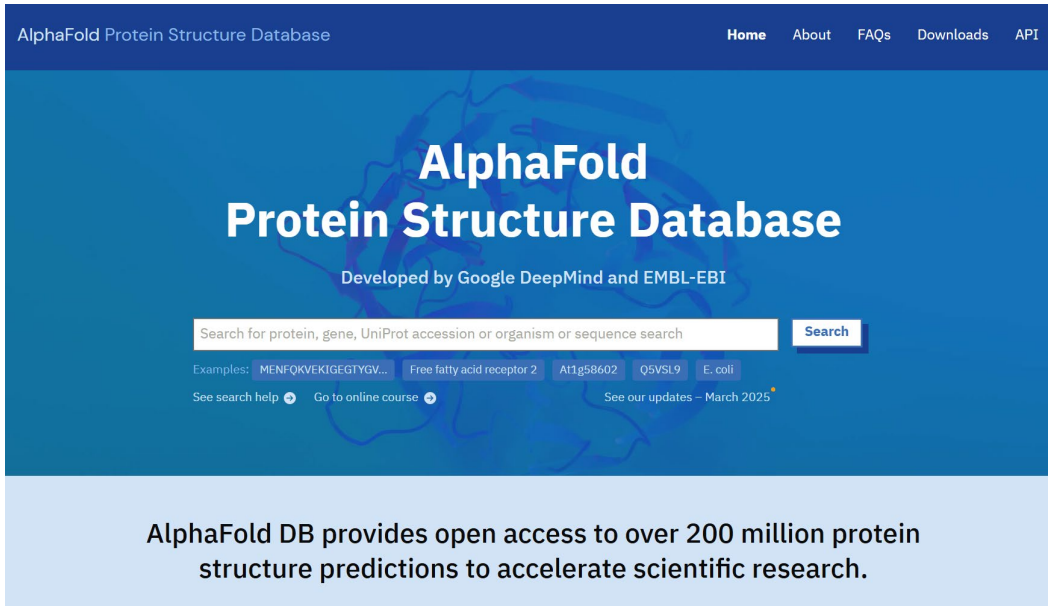
Special Topics: AI for Chemical Engineers

Lecture 2: Demonstration of Latest AI/ML Technology

University at Buffalo
The State University of New York

AlphaFold: an AI System that Predicts Protein 3D Structures from Amino Acid Sequences

- **AlphaFold 1 (2018)**: Introduced AI-based protein structure prediction; improved speed and accuracy.
- **AlphaFold 2 (2020)**: Major breakthrough with deep learning integration; achieved near-experimental accuracy; outperformed all competitors.
- **AlphaFold 3 (2024):** Expanding capabilities to protein-ligand interactions, RNA folding, and broader applications.



Try it out:

AlphaFold Server

Hemoglobin subunit beta (HBB):

MVHLTPEEKSAVTALWGKVNVDEV
GGEALGRLLVVYPWTQRFFESFGD
LSTPDAVMGNPKVKAHGKKVLGAF
SDGLAHLDNLKGTFAALSELHAHKL
RVDPVNFKLLSHCLLVTLAAHLPAE
FTPAVHASLDKFLASVSTVLTSKYR

https://alphafold.ebi.ac.uk/

Outline

- LLMs for Text & Reasoning
- Generative Media
- Vibe Coding
- Agentic AI
- Other AI/ML Tools

Limitation of LLMs in Early Years

Early version of chatbots (e.g., ChatGPT in 2023) suffered from:

- Short memory
- Hallucination
- Knowledge cutoff
- Short answers & slow speed
- Low intelligence
- Single input & output types

LLM Demonstration

Two years later, significant advances have been made to overcome above limitations. Here we will try:

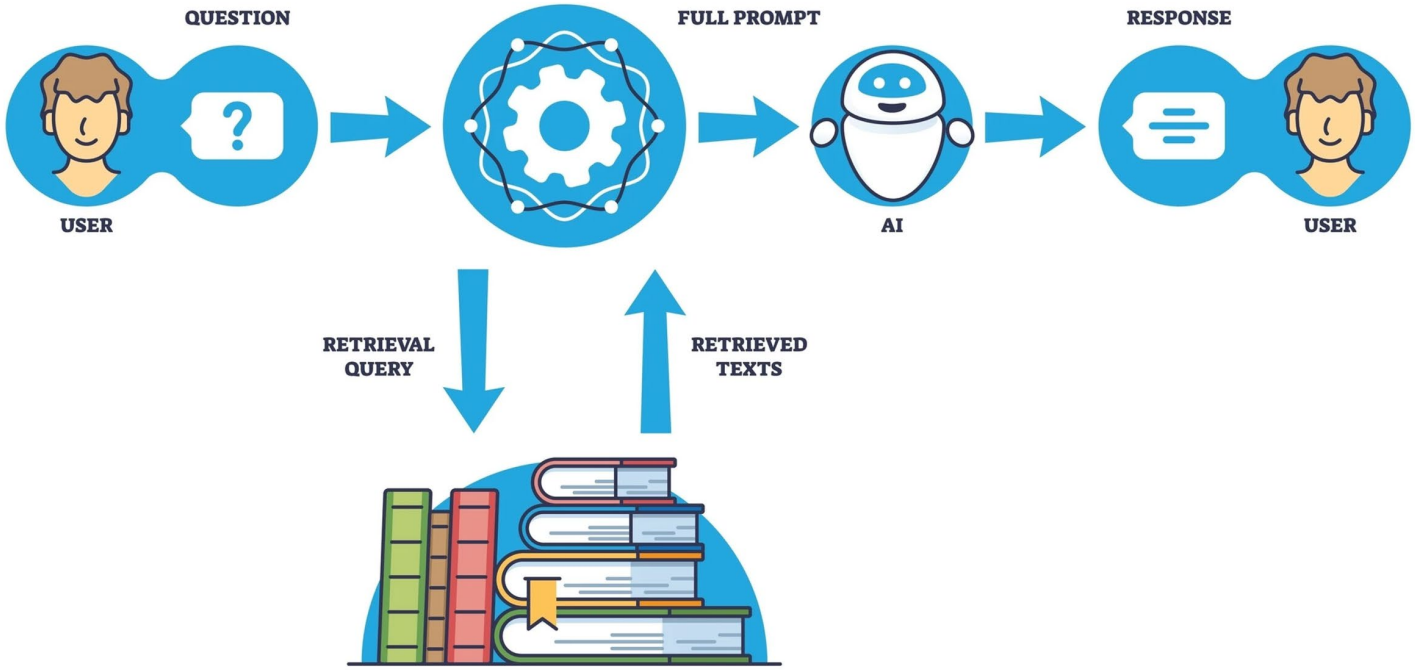
- Foundational LLMs (ChatGPT, Claude, Perplexity, Gemini, Grok, DeepSeek etc.); Homework set 1
  - Search literature; multi-modal prompting; deep research, data analysis
- Specialized LLMs (GPTZero, FutureHouse, GitHub Co-pilot, Google Learn About, etc.)
- You can now develop your personalized/customized ChatGPT (kind-of)

More readings about latest applications of LLM in science and engineering:

Zimmermann, Yoel, et al. "Reflections from the 2024 large language model (llm) hackathon for applications in materials science and chemistry." arXiv preprint arXiv:2411.15221 (2024).

Hands-on: Retrieval-Augmented Generation

Illustration of retrieval-augmented generation (RAG)



<u>Task:</u>
Try ChatPDF website to talk with your document.

Generative Media

LLM is multi-modal now, it can not only generate texts but can also generate media:

- Image generation (ChatGPT, Adobe Firefly, Google AI Studio, MidJourney etc.)
  - Prompt to try "*Generate an infographic showing a futuristic AI-powered chemical plant with robots monitoring distillation columns*"
  - Iterative image editing, ChatGPT edit selection tool, and modify a given image
- Sound generation (ElevenLabs, Google NotebookLM, Live AI assistant etc.)
  - Text to speech, voice changer, ChatGPT live AI assistant (multi-modal) etc.
  - Student's hands-on with NotebookLM.
- Video generation (Sora, Google Flow etc.)
  - Prompt to try "*Create a video of water molecules diffusing inside a nanostructured material*."
  - Limitation (consistency, realism, duration, "deepfake" concerns etc.) and current research frontier.

Vibe Coding

Vibe coding (term coined in early 2025) is an emerging software development practice that uses artificial intelligence (AI) to generate functional code from natural language prompts, accelerating development, and making app building more accessible, especially for those with limited programming experience.

- **“Pure” vibe coding**: In its most exploratory form, a user might fully trust the AI’s output to work as intended. The speed is the primary goal.
- **Responsible AI-assisted development**: This is the practical and professional application of the concept. In this model, AI tools act as a powerful collaborator or “pair programmer.” The user guides the AI but then reviews, tests, and understands the code it generates, taking full ownership of the final product.

https://cloud.google.com/discover/what-is-vibe-coding

Vibe Coding Demonstration

Pure vibe coding, for example:

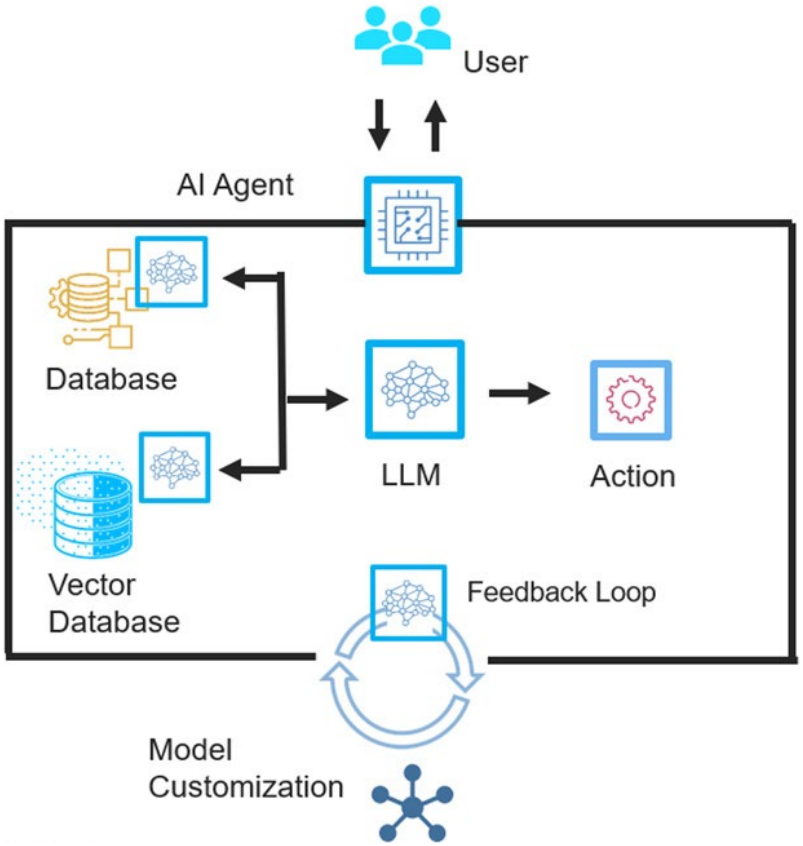
- Claude Artifacts (https://claude.ai/artifacts)
- Gemini (Build | Google AI Studio)
- And many more!

Responsible vibe coding, for example:

- GitHub copilot (in VS Code etc.)
- Google Colab (*Pro version free to students*)
- General LLMs (e.g., ChatGPT, Claude etc.)

What is Agentic AI?

Agentic AI uses sophisticated reasoning and iterative planning to autonomously solve complex, multi-step problems.



- **Perceive**: AI agents gather and process data from various sources, such as sensors, databases and digital interfaces.
- **Reason**: A large language model (LLM) acts as the orchestrator, or reasoning engine, that understands tasks, generates solutions and coordinates specialized models for specific functions like content creation, visual processing or recommendation systems.
- **Act**: By integrating with external tools and software via application programming interfaces, agentic AI can quickly execute tasks based on the plans it has formulated.
- **Learn**: Agentic AI continuously improves through a feedback loop, or “data flywheel,” where the data generated from its interactions is fed into the system to enhance models.

https://blogs.nvidia.com/blog/what-is-agentic-ai/
https://www.cadence.com/en_US/home/explore/agentic-ai.html

Agentic AI Examples

A few examples of agentic AI that is already available or under development:

- Autonomous customer service bots - Handle complex multi-turn conversations, access databases, process refunds, and escalate issues without human agents
- AI trading systems and financial advisors - Make investment decisions, rebalance portfolios, and execute trades based on market analysis and predefined strategies
- AI assistant for shopping, booking hotel, scheduling etc.
- Agentic AI for scientific discovery (AI-powered robotic agents for large-scale chemical testing, the AI scientist etc.)

The AI Scientist from Sakana AI (2024)

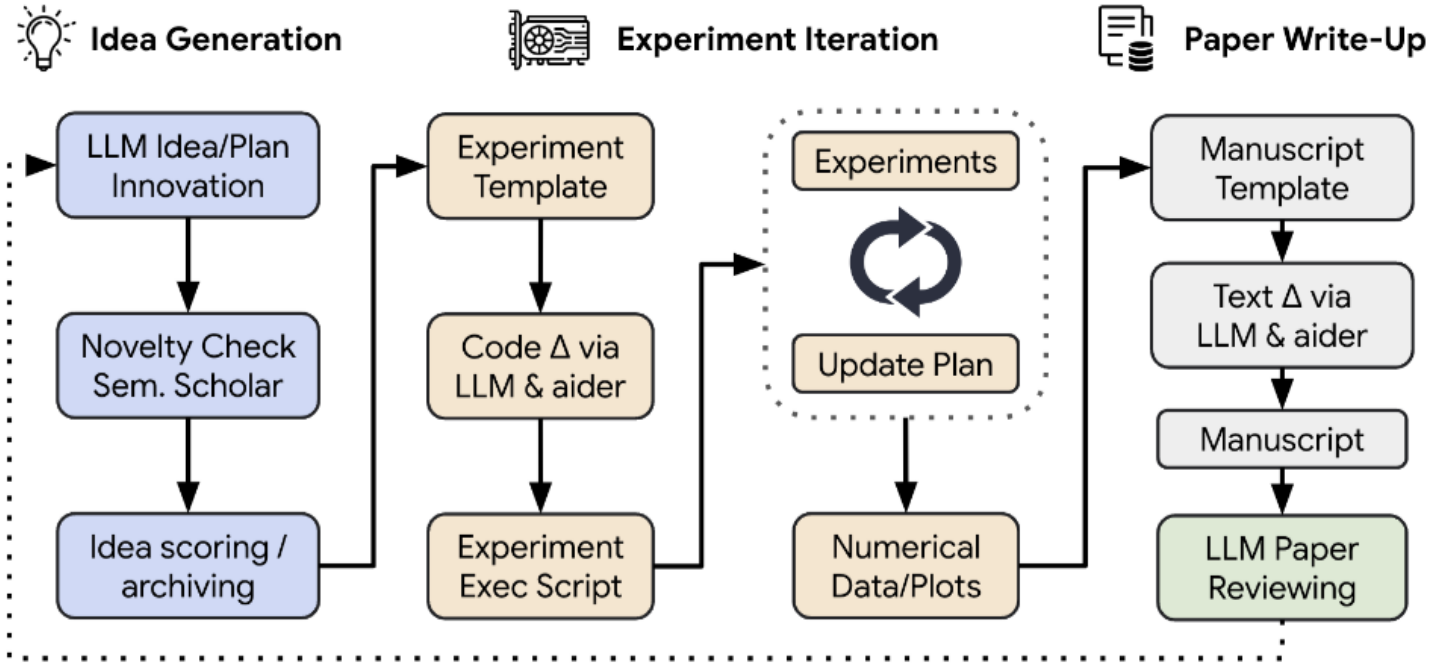


Figure 1 | Conceptual illustration of THE AI SCIENTIST, an end-to-end LLM-driven scientific discovery process. THE AI SCIENTIST first invents and assesses the novelty of a set of ideas. It then determines how to test the hypotheses, including writing the necessary code by editing a codebase powered by recent advances in automated code generation. Afterward, the experiments are automatically executed to collect a set of results consisting of both numerical scores and visual summaries (e.g. plots or tables). The results are motivated, explained, and summarized in a LaTeX report. Finally, THE AI SCIENTIST generates an automated review, according to current practice at standard machine learning conferences. The review can be used to either improve the project or as feedback to future generations for open-ended scientific discovery.

https://arxiv.org/pdf/2408.06292
https://github.com/SakanaAI/AI-Scientist

Agentic AI Demonstration

"Preliminary" version:

- ChatGPT plugins

Recent advancements:

- ChatGPT Agent mode
- GitHub Copilot Agent mode

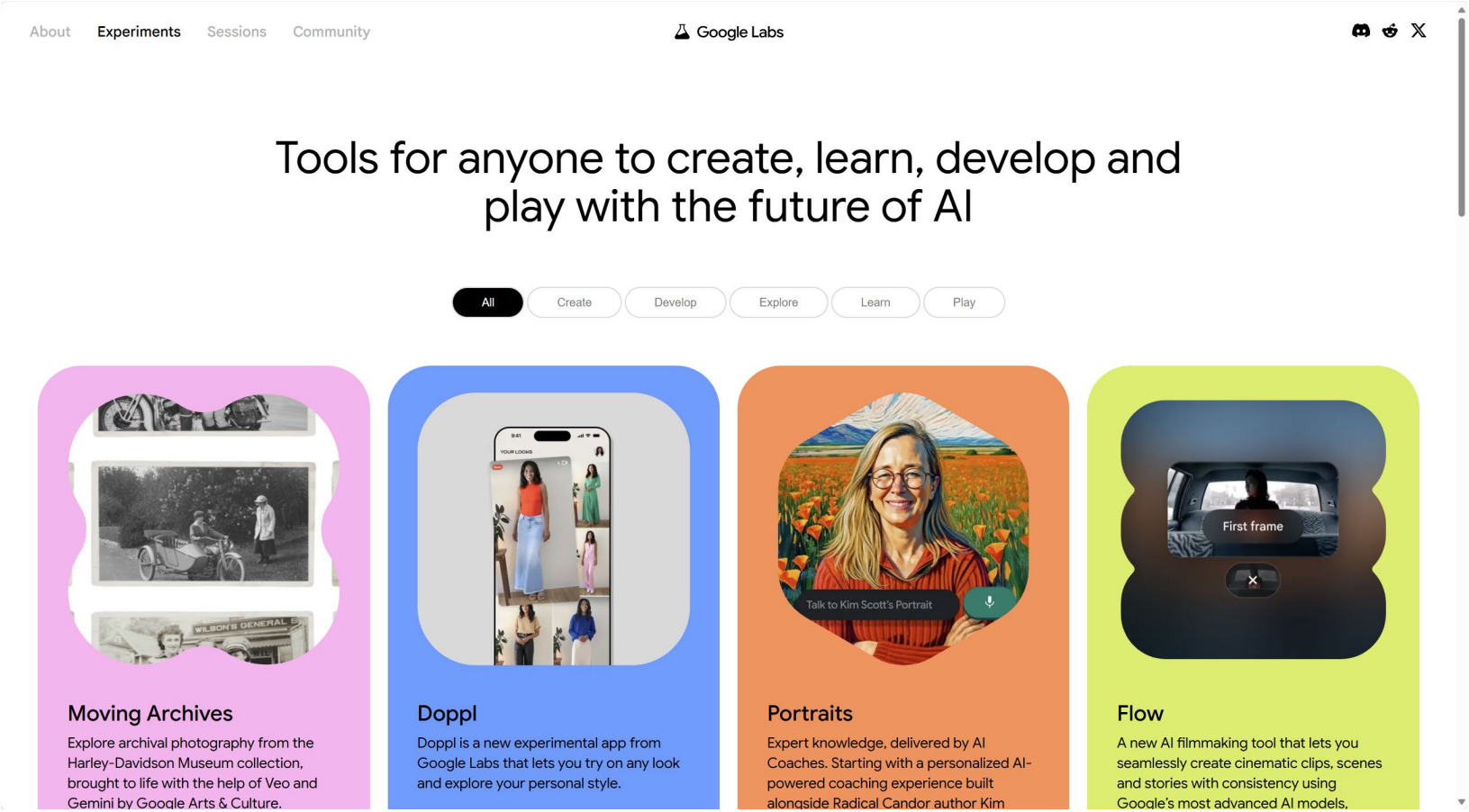
Hands-on (free trial):

- Manus AI (https://manus.im/app)

Build your own AI agent:

- The AI scientist (https://github.com/SakanaAI/AI-Scientist)
- AutoGen (https://github.com/microsoft/autogen)
- LangChain (https://github.com/langchain-ai/langchain)

Other AI/ML Tools



Many fun AI tools from Google Labs (https://labs.google/experiments). Try them out!

Open Discussions

- How do you think these AI capabilities might change what skills are most valuable for chemical engineers in 5-10 years? What should you focus on learning that AI can't replace?